

SIMON NDUATI NGIGI

Nairobi, Kenya

Email — LinkedIn — GitHub — Portfolio

PROFESSIONAL SUMMARY

Data Scientist with hands on experience building production data pipelines, automation systems, and analytics tooling: from OCR driven document processing to multi region reporting infrastructure and compliance alerting. Skilled in Python, SQL, and pipeline design, with a focus on data integrity, catching silent errors, reconciling conflicting sources, and building systems that hold up under daily use. Also independently designed and deployed end to end data and ML applications on self sourced datasets.

EDUCATION

Zetech University, Nairobi, Kenya
Bachelor of Science in Software Engineering

Expected Dec 2026

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, C

Data & Machine Learning: Pandas, NumPy, Scikit-Learn, Random Forest, LSTM, ANN, EDA

Data Engineering: ETL Pipelines, PostgreSQL, Supabase, Web Scraping (BeautifulSoup, Playwright)

Automation & Backend: n8n, FastAPI, Next.js, Docker, WhatsApp/Evolution API

Visualization: Plotly, Dash, Streamlit

Tools: Git, GitHub, Linux, VS Code

PRODUCTION SYSTEMS — FIRESIDE GROUP

Data Analyst, Analytics Systems
Fireside Group

Fuel Extractor System: OCR Pipeline and WhatsApp Integration

PaddleOCR, Python, Supabase, WhatsApp API

- Built an end to end fuel and maintenance system where drivers submit receipts via WhatsApp and PaddleOCR automatically extracts structured data into Supabase.
- Eliminated manual receipt entry and enabled real time odometer tracking across the vehicle fleet.

Fireside PI-DB: Secure Staff Directory and Audit Compliance Portal

Next.js, Supabase, PostgreSQL

- Built a full stack internal system for managing employee records, department assignments, and vehicle allocations, with automatic audit logging on every change.
- Replaced fragmented spreadsheets with a single auditable source of truth and enforced rules preventing invalid states.

Multi Region Fleet Reporting and Driver Behaviour Automation

n8n, Python, Playwright, FastAPI, Docker

- Built an automation pipeline that logs into the fleet tracking platform to generate motion reports and parses emailed driver behaviour spreadsheets across three regions.
- Replaced a fully manual, region by region spreadsheet process with standardized, database ready ingestion.

Vehicle Compliance Alert System: Insurance and Inspection Tracking

Node.js, node-cron, Google Sheets API

- Built an automated compliance system monitoring insurance and inspection expiry across a 98 vehicle fleet, sending scheduled WhatsApp alerts to logistics stakeholders.
- Replaced reactive discovery of lapsed documents with proactive, automatic alerting.

INDEPENDENT PROJECTS

Kenya Property Price Predictor

Python, Scikit-Learn, Streamlit

- Built a proprietary real estate dataset by scraping and cleaning 1,733 live property listings from Kenyan platforms, replacing reliance on generic static datasets.
- Engineered a Random Forest regression model achieving an R^2 score of 0.683, then deployed it as a live app that reduces property valuation time from hours to under 5 seconds.

M-Pesa Financial Analysis Dashboard

Python, Power BI, Plotly

- Extracted and structured 96 months of Central Bank of Kenya financial reports into a clean time series dataset spanning a 7 year period.
- Built an interactive dashboard visualizing transaction volume trends and agent growth, eliminating manual report analysis and speeding up financial insight generation.

Kenya News Sentiment Tracker

NLP, TextBlob, Web Scraping

- Engineered a real time NLP pipeline scraping and processing 200+ daily news headlines from Standard Media across six categories.
- Automated sentiment analysis to quantify media tone, replacing manual monitoring with a continuously updating live dashboard.

NSE Stock Tracker Dashboard

Python, Plotly, Web Scraping

- Built a real time data pipeline scraping live stock prices for major NSE listed companies.
- Developed an interactive dashboard for monitoring price movements, volatility, and trading trends, reducing reliance on manual market tracking.

Stock Prediction Dashboard

LSTM, Deep Learning, Dash

- Built an LSTM based deep learning model forecasting stock price trends from historical time series data.
- Deployed the model as an interactive Dash web application for exploring predictions in real time.

Heart Disease Classification

ANN, Scikit-Learn

- Developed an Artificial Neural Network classifying cardiovascular risk from patient health metrics.
- Demonstrated proficiency in structured medical data modelling within a HealthTech context.

OTHER EXPERIENCE

Data Science Virtual Intern

CodSoft International (Remote)

- Developed and evaluated machine learning models for classification and prediction tasks, improving feature selection through structured exploratory data analysis.
- Optimized ETL pipelines and preprocessing workflows, enhancing model performance and generalization across multiple datasets.